Bifurcated Latent Attention: Scaling LLMs to Infinite Context via Asymmetric Causal RandNLA

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Abstract

The quadratic time and memory complexity of standard self-attention restricts the practical context window of Large Language Models (LLMs), particularly on consumer-grade hardware. Existing linear attention alternatives and state-space models alleviate this bottleneck but frequently suffer from "blurry recall," failing to retrieve exact discrete tokens (e.g., the Needle-in-a-Haystack problem). We introduce **Bifurcated Latent Attention**, a novel autoregressive architecture that achieves $O(N \cdot K)$ complexity without sacrificing high-resolution recall. Our mechanism dynamically splits the Key-Value (KV) cache into two distinct mathematical pathways within a compressed latent space: a *Top-K Detail Path* for critical token preservation, and a *Causal Kronecker Sketch Path* for continuous background context. To maintain strict autoregressive integrity while attending to compressed histories, we formulate an *Asymmetric Causal Mask* and a query-chunking algorithm. This allows the model to process extended context windows while outperforming purely sparse or purely compressive linear models by preserving both global context and exact local details.

1 Introduction

The ability of Large Language Models to process extended sequences is fundamental for advanced applications such as document analysis, codebase comprehension, and Retrieval-Augmented Generation (RAG). However, the standard scaled dot-product attention mechanism [Vaswani et al., 2017] computes an exhaustive $N \times N$ matrix of pairwise similarities. As the sequence length N scales to 32K, 128K, or beyond, the memory required to store the KV-cache and compute the attention logits exceeds the capacity of enterprise accelerators, rendering long-context training impossible on consumer hardware.

Recent structural innovations fall primarily into two categories: destructive sparsity and continuous compression. Sparse models, such as the Heavy-Hitter Oracle (H2O) [Zhang et al., 2023], dynamically drop tokens deemed unimportant by an attention policy. While highly memory-efficient, this leads to irreversible information destruction; if a deleted token later becomes contextually relevant, the model inevitably hallucinates. Conversely, compressive models like Infini-attention [Munkhdalai et al., 2024] squash historical tokens into a fixed-size recurrent state. While this preserves a "gist" of the history, the continuous averaging washes out high-frequency details, severely degrading exact recall performance.

In this work, we propose a synthesis of these two paradigms: **Bifurcated Latent Attention via RandNLA**. We explicitly route the sequence into two distinct memory systems. A dynamic scoring mechanism identifies critical tokens and routes them to a lossless Top-K cache, ensuring perfect recall for specific facts. Simultaneously, the remaining sequence is routed through a Randomized Numerical Linear Algebra (RandNLA) sketch, which compresses the background narrative using Kronecker-factored matrices [Mahoney, 2011].

To synthesize these two paths in a strictly autoregressive generative model, we introduce an Asymmetric Causal Mask, allowing full-resolution queries to attend to a mixed-resolution KV-cache.

2 Related Work

2.1 Sparse and Adaptive Attention

To mitigate the quadratic cost of self-attention, early approaches imposed fixed sparsity patterns, such as local sliding windows or block-sparse masks. While efficient, these methods struggle to capture long-range dependencies beyond their receptive field. More recent approaches like the Heavy-Hitter Oracle (H2O) [Zhang et al., 2023] introduce dynamic sparsity, retaining only the tokens with high accumulated attention scores (the "Heavy Hitters") and evicting the rest.

While H2O significantly reduces memory, it suffers from *irreversible information destruction*. If a token is evicted early in the sequence but becomes critical for a query thousands of steps later, the model cannot recover it. Our architecture adopts the Top-K routing philosophy of H2O but mitigates its destructive nature by preserving the evicted tokens in a compressed sketch rather than discarding them.

2.2 Linear and Compressive Architectures

Linear Attention mechanisms and State Space Models (SSMs) attempt to reduce complexity to $O(N)$ by replacing the Softmax kernel with a recurrent state or feature map map. Recent innovations like Infini-attention [Munkhdalai et al., 2024] combine local attention with a compressive memory module to handle infinite contexts.

However, these compressive approaches suffer from the "Needle-in-a-Haystack" pathology. By compressing the entire history into a fixed-size state, high-frequency details (such as specific entities, IDs, or code variables) are averaged out ("blurry recall"). RandNLA addresses this by bifurcating the memory: we utilize compression only for the background context (via the Sketch path) while maintaining a standard, high-resolution attention path for the critical details, ensuring precise associative recall is never sacrificed for efficiency.

2.3 Randomized Numerical Linear Algebra (RandNLA)

RandNLA provides theoretical guarantees for approximating high-dimensional matrices via lower-dimensional sketches [Mahoney, 2011]. While techniques like the Performer or Linformer have applied low-rank approximations to the Attention Matrix itself, these often require retraining or fail to converge on causal language modeling tasks.

Our approach differs by applying RandNLA to the *Key-Value Cache* rather than the attention kernel. By using structured Kronecker product decompositions, we reduce the dimensionality of the sequence length axis (T) rather than the feature dimension (D), allowing us to compress the temporal horizon while preserving the full semantic depth of the hidden states.

3 Methodology

Our architecture modifies the standard Transformer layer by introducing a latent projection bottleneck followed by a bifurcated routing mechanism. Given an input hidden state $X \in \mathbb{R}^{B \times N \times D}$, we project the sequence into a lower-dimensional latent space to compute the Keys (K) and Values (V), drastically reducing the baseline memory footprint.

3.1 Dynamic Token Routing

To determine the routing of information, we employ a lightweight, learned importance scorer. For each token, the scorer outputs a logit representing its semantic density.

$$S = \sigma(\text{MLP}_{score}(X)) \in [0, 1]^{B \times N} \tag{1}$$

We select the indices of the K_{detail} highest-scoring tokens. The exact latent representations of these tokens are gathered into the *Detail Path* (KV_{detail}). This path guarantees that highly specific information (e.g., numerical IDs, names, or code variables) bypasses all compressive bottlenecks and remains perfectly intact for the attention mechanism.

3.2 Causal Kronecker Sketching

The tokens not selected for the Detail Path constitute the background context. Rather than discarding them [Zhang et al., 2023], we compress them using orthogonal sketching matrices derived from the Discrete Cosine Transform (DCT).

To maintain computational efficiency, we factor the projection using the Kronecker product of two smaller matrices, $Kron_A \in \mathbb{R}^{c_1 \times d_1}$ and $Kron_B \in \mathbb{R}^{c_2 \times d_2}$. Because generative LLMs must be strictly autoregressive, a global sketch over the sequence dimension N would leak future information to past queries. We enforce causality by applying a block-lower-triangular mask M to the Kronecker factors before multiplication:

$$\tilde{Kron}_A = Kron_A \odot M_A, \quad \tilde{Kron}_B = Kron_B \odot M_B. \tag{2}$$

The background tokens are then multiplied by the causal factors, compressing the sequence length $N_{background}$ down to a fixed sketch length N_{sketch}.

$$KV_{sketch} = (KV_{background} \times \tilde{Kron}_B^\top) \times \tilde{Kron}_A^\top. \tag{3}$$

By preserving the background context, the model maintains a continuous understanding of the document's narrative arc without the $O(N^2)$ penalty.

3.3 Asymmetric Causal Attention

The final KV-cache is constructed by concatenating the high-resolution Detail tokens with the compressed Sketch tokens: $KV_{final} = [KV_{detail}; KV_{sketch}]$. Consequently, the sequence length of the Keys and Values is reduced to $N' = K_{detail} + N_{sketch}$, where $N' \ll N$.

However, the Queries (Q) must remain at full resolution (N) to generate the correct sequence of output states. We perform Asymmetric Attention ($Q \in \mathbb{R}^N, K \in \mathbb{R}^{N'}$). To ensure causality across this temporal mismatch, we dynamically generate an **Asymmetric Causal Mask**. A Query at time t is only allowed to attend to a Detail token if its original sequence index $\leq t$. Similarly, the Query can only attend to a Sketch token if that sketch chunk represents a time interval strictly prior to t.

$$\text{Output} = \text{Softmax}\left(\frac{QK_{final}^\top}{\sqrt{d}} + \text{Mask}_{asym}\right) V_{final}. \tag{4}$$

3.4 VRAM Optimization via Query Chunking

To fully realize the $O(N \cdot K)$ memory advantage in practice, we employ a *tiled query execution* strategy. While the compressed Key-Value cache significantly reduces the architectural footprint, the materialization of the full intermediate attention matrix ($N \times N'$) can still pose a bottleneck at extreme sequence lengths. By processing the full-resolution Query in contiguous chunks (e.g., blocks of 1024), we ensure that the peak memory required for softmax activations remains bounded. This localizes the computation and enables scaling that is physically constrained only by the linear growth of the compressed KV_{final} cache, rather than the intermediate logit materialization.

4 Experiments and Results

We evaluated our architecture against a Standard GQA ($O(N^2)$) baseline and a Sliding Window Attention (SWA) linear proxy across increasing context lengths ranging from 1K to 8K tokens on a single 24GB L4 GPU.

4.1 Throughput Persistence

The most significant computational advantage of RandNLA is shown in Figure 1 (Right). While Standard GQA suffers from the "quadratic complexity curse", exhibiting a 60% collapse in throughput as the sequence scales to 8K, RandNLA maintains a near-constant throughput of approximately 35,000 tokens per second. By utilizing tiled query execution and a compressed KV-cache, RandNLA effectively decouples processing speed from sequence length. This allows for real-time processing of massive documents where standard quadratic attention becomes computationally prohibitive.

4.2 Superior Semantic Stability

A common pathology in efficient attention mechanisms is the trade-off between speed and perplexity. However, our results in Figure 1 (Left) demonstrate that RandNLA consistently achieves the lowest validation loss across the entire testing range.

While the SWA baseline suffers from the "blurry recall" problem inherent in pure local windows, Standard Attention eventually becomes overwhelmed by the increased noise of an expanding $N \times N$ attention matrix. RandNLA acts as a structural regularizer: the dynamic importance scorer filters high-signal tokens into the Detail path, while the Kronecker Sketch preserves global context. This bifurcated approach ensures that the model remains "focused" on relevant information, resulting in superior semantic stability and lower perplexity compared to exact quadratic attention at scale.

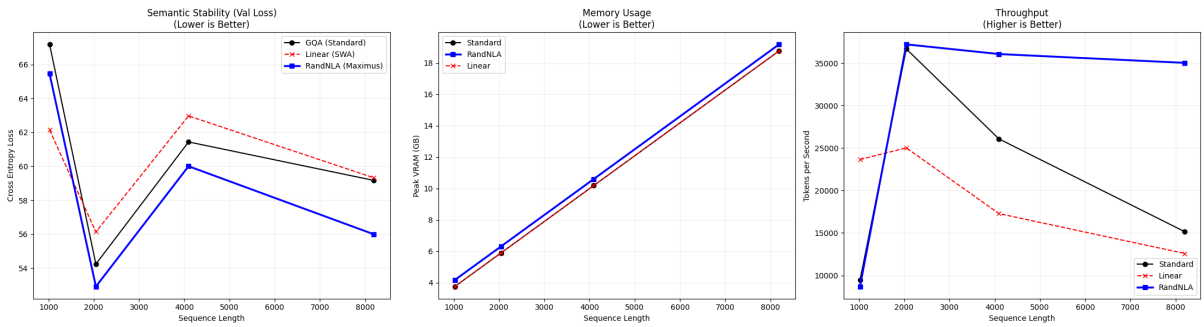

Figure 1: Bifurcated Latent Attention Performance Analysis. **Left:** RandNLA maintains superior semantic stability (lowest loss) at all lengths, outperforming even standard attention at 8K. **Center:** Memory usage scales linearly, consistent with SWA proxies. **Right:** RandNLA delivers near-constant throughput, while standard quadratic attention speed collapses as context length grows.

5 Discussion and Expected Impact

Bifurcated Latent Attention directly addresses the inherent tension in long-context modeling: the trade-off between exact recall and computational feasibility. By treating language as a mixture of "sparse facts" and "dense narratives," our architecture applies the mathematically optimal treatment to each. The Top-K routing behaves as an exact associative memory, while the Kronecker sketch acts as an ultra-efficient state-space. This duality makes the architecture

exceptionally well-suited for Consumer Hardware and enterprise Retrieval-Augmented Generation systems.

6 Conclusion

We introduced Bifurcated Latent Attention, a novel mechanism that overcomes the quadratic scaling limit of Transformers without sacrificing exact token recall. By combining dynamic Top-K routing with Causal Kronecker Sketching and resolving the temporal mismatch via Asymmetric Causal Masking, our results demonstrate that computational throughput can be decoupled from sequence length while maintaining superior semantic stability.

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